

Part 3: Engineering Plan & Proof of Concept

Part 1 delivers a working single-URL crawler deployed on GCP Cloud Run. Part 3 describes how to scale it into the distributed system designed in Part 2.

Implementation Plan

Phase 1: Core Crawl Pipeline (1 Week)

Build:

- URL ingestion service (read from GCS file or Cloud SQL, normalize, deduplicate via Redis Bloom filter, publish to Pub/Sub)
- Crawler workers on GKE subscribing to Pub/Sub crawl topic
- Shared per-domain rate limiting via Redis so all worker pods see the same counter per domain
- Raw content upload to GCS on successful crawl

Deliverable: Distributed crawling functional. URLs flow from input file through the queue into GKE workers, raw HTML lands in GCS.

Phase 2: Extraction Pipeline (1 Week)

Build:

- Redis write buffer + DB flusher for Cloud SQL writes
- Content hash comparison to skip unchanged pages
- Extraction workers on GKE subscribing to extraction topic
- Batch KeyBERT topic extraction
- Bulk write to Elasticsearch

Deliverable: End-to-end pipeline functional. Crawled pages are parsed, classified, topics extracted, and searchable in Elasticsearch.

Phase 3: Reliability & Observability (1 Week)

Build:

- Dead Letter Topic after 5 retries with structured error metadata
- Cloud Monitoring dashboards (queue backlog, throughput, error rate, P95 latency)

- Alerting on all SLO thresholds defined in Part 2
- Stats table queries surfaced as internal dashboards for per-domain visibility

Deliverable: Production-operable pipeline. Failures are captured, metrics are visible, alerts fire on degradation.

Phase 4: Scale Validation & Load Testing (1 Week)

Load test at 10% scale (100M URLs), then full 1B:

- Measure actual crawl and extraction latency under sustained GKE load (the 1s/3s estimates from Part 2 are based on local testing and need to be validated)
- Sustained throughput at target URLs/sec for 1+ hour
- Queue lag under load (Pub/Sub oldest unacked message age)
- CPU and memory utilisation per instance type — validate or revise instance count estimates from Part 2
- Failure handling end-to-end: confirm Dead Letter Topic captures failures and retry recovers correctly

Deliverable: Load test report with actual throughput, latency, and resource utilisation figures. Revised instance count estimates if actual numbers differ from Part 2. Go/no-go recommendation for production.

Potential Blockers

Things to monitor during Phase 4

- **Playwright memory behaviour:** Chromium's memory usage under hours of sustained concurrent crawling is unknown. During load testing we need to track per-pod memory over time. If leaks are observed, a browser restart policy (recycle after N pages or M minutes) would be the fix, but we won't know if this is needed until we run it.
- **Actual latency numbers:** the 1s crawl / 3s extraction estimates come from local single-instance testing. Real behaviour under sustained GKE load needs to be measured in Phase 4.

Genuine unknowns

- **IP reputation blocking:** whether GKE egress IPs are blocked by bot-protection systems on target domains won't be known until the crawl runs. Residential proxy integration is the reliable fix but adds cost and vendor dependency. This is a product decision as much as an engineering one. Timeline: 1–2 weeks if the proxy route is taken.
-

PoC Success Criteria

Run of at least 10 million URLs:

Criterion	Pass Threshold
Crawl throughput	≥ 1,000 URLs/sec sustained for 1 hour
Success rate	≥ 90% reach success or blocked
Data completeness	< 5% of successful crawls have null title
Topic extraction	≥ 80% of successful crawls produce ≥ 1 topic
End-to-end latency P95	≤ 30 minutes

Release Plan

Alpha: Internal (1M URLs)

- Controlled test set covering all page types
- Review of metadata and topic quality
- Gate: all PoC criteria pass

Beta: Limited Production (100M URLs)

- Production domain lists
- Gate: throughput ≥ 3,000 URLs/sec, error rate < 5%, no data loss

General Availability

- Full 5,000 URLs/sec, all Part 2 SLAs enforced
- Runbooks rehearsed, DR tested
- Gate: two consecutive clean beta runs with no P1 incidents